WORKED SOLUTION

STEP I 1995, Question 7

Pure • Vector Geometry of Triangles

STEP I 1995, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s1-q7

The key idea

Putting the origin at the point equidistant from A, B and C means the position vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ all have the same length. That one fact does all the work, because for vectors of equal length $(\mathbf{b}+\mathbf{c})\cdot(\mathbf{c}-\mathbf{b}) = |\mathbf{c}|^2 - |\mathbf{b}|^2 = 0$. Once we find that Q has the beautifully simple position vector $\mathbf{a}+\mathbf{b}+\mathbf{c}$, both parts become a line or two of algebra.

Solution

Choosing the origin

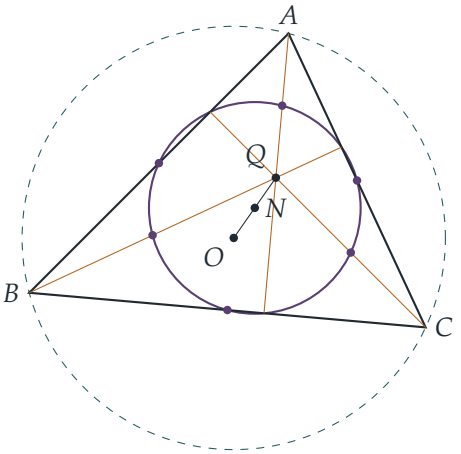
The points equidistant from A and B form the perpendicular bisector of AB, a straight line perpendicular to AB; similarly for B and C. Since A, B, C are not collinear, AB and BC are not parallel, so neither are their perpendicular bisectors. These two lines therefore meet in exactly one point, which is equidistant from A and B and from B and C, and so from all three. (It is the circumcentre of the triangle.)

Take this point as O, write $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ for the position vectors of A, B, C, and let R be the common distance, so

$$|\mathbf{a}| = |\mathbf{b}| = |\mathbf{c}| = R.$$

The centroid has position vector $\mathbf{g} = \frac{1}{3}(\mathbf{a}+\mathbf{b}+\mathbf{c})$. Since $\overrightarrow{GQ} = 2\overrightarrow{OG}$ means $\mathbf{q}-\mathbf{g} = 2\mathbf{g}$,

$$\mathbf{q} = 3\mathbf{g} = \mathbf{a}+\mathbf{b}+\mathbf{c}, \qquad \mathbf{n} = \tfrac{1}{2}\mathbf{q} = \tfrac{1}{2}(\mathbf{a}+\mathbf{b}+\mathbf{c}).$$



The figure shows the circle through A, B, C centred at O (dashed), the altitudes meeting at Q, and the circle of radius $\frac{1}{2}R$ centred at N through the six midpoints of part (ii).

Part (i)

We have $\overrightarrow{AQ} = \mathbf{q} - \mathbf{a} = \mathbf{b} + \mathbf{c}$ and $\overrightarrow{BC} = \mathbf{c} - \mathbf{b}$, so

$$\overrightarrow{AQ} \cdot \overrightarrow{BC} = (\mathbf{c} + \mathbf{b}) \cdot (\mathbf{c} - \mathbf{b}) = |\mathbf{c}|^2 - |\mathbf{b}|^2 = R^2 - R^2 = 0,$$

using $\mathbf{b} \cdot \mathbf{c} = \mathbf{c} \cdot \mathbf{b}$ to cancel the cross terms. Hence $\overrightarrow{AQ}$ is perpendicular to $\overrightarrow{BC}$.

? **Ask yourself** Does $\overrightarrow{AQ} \cdot \overrightarrow{BC} = 0$ really put Q on the altitude from A? What if $\overrightarrow{AQ}$ is the zero vector?

If $Q \neq A$, the line AQ passes through A and is perpendicular to BC, so it *is* the altitude from A, and Q lies on it. If $Q = A$ (which happens exactly when $\mathbf{b} + \mathbf{c} = \mathbf{0}$, i.e. BC is a diameter and the angle at A is a right angle), then Q lies on the altitude from A trivially. Either way, Q is on the altitude from A.

Nothing in the argument singled out A: the expression $\mathbf{a} + \mathbf{b} + \mathbf{c}$ is symmetric, so in the same way

$$\overrightarrow{BQ} \cdot \overrightarrow{CA} = (\mathbf{c} + \mathbf{a}) \cdot (\mathbf{a} - \mathbf{c}) = |\mathbf{a}|^2 - |\mathbf{c}|^2 = 0,$$
$$\overrightarrow{CQ} \cdot \overrightarrow{AB} = (\mathbf{a} + \mathbf{b}) \cdot (\mathbf{b} - \mathbf{a}) = |\mathbf{b}|^2 - |\mathbf{a}|^2 = 0,$$

and Q also lies on the altitudes from B and from C. So all three altitudes pass through the single point Q: they are concurrent. (Q is the orthocentre.)

Part (ii)

The midpoint of AQ has position vector $\frac{1}{2}(\mathbf{a} + \mathbf{q}) = \mathbf{a} + \frac{1}{2}(\mathbf{b} + \mathbf{c})$, so its displacement from N is

$$\mathbf{a} + \tfrac{1}{2}(\mathbf{b} + \mathbf{c}) - \tfrac{1}{2}(\mathbf{a} + \mathbf{b} + \mathbf{c}) = \tfrac{1}{2}\mathbf{a}.$$

The midpoint of BC has position vector $\frac{1}{2}(\mathbf{b}+\mathbf{c})$, so its displacement from N is

$$\tfrac{1}{2}(\mathbf{b}+\mathbf{c}) - \tfrac{1}{2}(\mathbf{a}+\mathbf{b}+\mathbf{c}) = -\tfrac{1}{2}\mathbf{a}.$$

Both of these have length $\frac{1}{2}|\mathbf{a}| = \frac{1}{2}R$.

By the same symmetry, the midpoints of BQ and CA are displaced from N by $\pm\frac{1}{2}\mathbf{b}$, and the midpoints of CQ and AB by $\pm\frac{1}{2}\mathbf{c}$. Since $|\mathbf{a}| = |\mathbf{b}| = |\mathbf{c}| = R$, all six midpoints are at distance $\frac{1}{2}R$ from N, which is what we had to show. In fact we have shown a little more: the midpoint of AQ and the midpoint of BC are at opposite ends of a diameter of this circle, and likewise for the other two pairs.

Checks

- An equilateral triangle: then $\mathbf{a}+\mathbf{b}+\mathbf{c} = \mathbf{0}$, so O, G, Q and N all coincide, and the six midpoints should all be at distance $\frac{1}{2}R$ from the centre. Indeed the midpoints of the sides are at distance $R\cos 60^\circ = \frac{1}{2}R$, and the midpoints of AQ etc. are halfway along radii.
- A right angle at A: then $Q = A$, as the argument above predicts, since the two legs are themselves altitudes through A.
- Numerically, we took 10^5 random triangles, computed the circumcentre directly, and confirmed $Q = \mathbf{a}+\mathbf{b}+\mathbf{c}$, that all three dot products vanish, and that all six midpoints are at distance $\frac{1}{2}R$ from N, with largest relative error about 10^{-13}.

Common slips

- Reading $\overrightarrow{GQ} = 2\overrightarrow{OG}$ as $\mathbf{q} = 2\mathbf{g}$. It says $\mathbf{q}-\mathbf{g} = 2\mathbf{g}$, so $\mathbf{q} = 3\mathbf{g}$.
- Stopping after one perpendicularity. Concurrency needs Q on all three altitudes; say explicitly that the same calculation works for B and C.
- Forgetting to justify the origin: the reason it exists is that the perpendicular bisectors of two non-parallel sides must meet.

Beyond STEP

Since $\mathbf{q} = 3\mathbf{g}$ and $\mathbf{n} = \frac{3}{2}\mathbf{g}$, the points O, G, N, Q all lie on one line, the *Euler line*, with $OG : GQ = 1 : 2$. The circle of part (ii) is the *nine-point circle*: it also passes through the three feet of the altitudes (visible in the figure), because each foot sees a diameter such as the one from the midpoint of AQ to the midpoint of BC at a right angle. Choosing a clever origin so that a symmetric expression does the work is a recurring trick in vector geometry and in linear algebra; Strang's *Introduction to Linear Algebra* develops the vector and dot-product tools further, and Siklos's *Advanced Problems in Mathematics* has more STEP-style geometry in this spirit.

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